DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE; THE FACILITY OWNER'S QUALIFIED PROCEDURES AND APPLICABLE CODES GOVERN.

# WO-300-01 — Pull, clean and test the tube bundle of E-101

| | | | |
|---|---|---|---|
| **Area** | A300 Exchangers and drum | **Task** | XB Exchanger bundle pull, clean and tube-side test |
| **Priority** | 2 (integrity) | **Issued by** | Facility owner (demonstration) |
| **Equipment** | E-101 | **Lines** | 4"-P300-P-1003, 4"-P300-P-1005, 4"-P150-CWS-3001, 4"-P150-CWR-3002, 1"-P150-V-6021, 1"-P150-D-6022, 3/4"-P300-I-6023 |
| **Isometrics** | ISO-4"-P300-P-1003, ISO-4"-P300-P-1005, ISO-4"-P150-CWS-3001, ISO-4"-P150-CWR-3002, ISO-1"-P150-V-6021, ISO-1"-P150-D-6022, ISO-3/4"-P300-I-6023 | | |
| **Permits** | General work permit, Confined-space entry permit, Lifting plan | | |

## Reason

Fouling: tube bundle to be pulled, cleaned and tested.

## Scope of work

1. Isolate E-101 per the isolation list (tube and shell sides); drain and vent.
2. Remove the channel cover; pull the tube bundle to the cleaning bay.
3. Clean the tubes and shell side; inspect visually; thickness at the TMLs.
4. Reinstall the bundle with new gaskets; tube-side hydrostatic test at 420 psi (SW-TEST-1).
5. Remove blinds; restore the line-up.

Affected items: E-101

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-TEST-1, SW-WO-1, SW-WO-2, SW-WO-3, SW-WO-5, SW-WO-6

## Isolation list (SW-WO-3)

| # | Kind | Tag | Line | Action | Location | Fitted ✓ | Removed ✓ |
|---|---|---|---|---|---|---|---|
| 1 | valve | 4"-P300-P-1003/B2 | 4"-P300-P-1003 | Close and lock VALVE, GATE, WEDGE, OS&Y 4" P300 (4"-P300-P-1003/B2) on E-101-N1 | E 110'-6" N -25'-0" EL 109'-3-13/16" | | |
| 2 | blind | E-101-N1 | 4"-P300-P-1003 | Fit a blind between 4"-P300-P-1003/B2 and the nozzle flange E-101-N1 | E 110'-6" N -25'-0" EL 108'-8-1/16" | | |
| 3 | valve | 4"-P300-P-1005/A2 | 4"-P300-P-1005 | Close and lock VALVE, GATE, WEDGE, OS&Y 4" P300 (4"-P300-P-1005/A2) on E-101-N2 | E 110'-6" N -25'-0" EL 103'-8-3/16" | | |
| 4 | blind | E-101-N2 | 4"-P300-P-1005 | Fit a blind between 4"-P300-P-1005/A2 and the nozzle flange E-101-N2 | E 110'-6" N -25'-0" EL 104'-3-15/16" | | |
| 5 | valve | 4"-P150-CWS-3001/B2 | 4"-P150-CWS-3001 | Close and lock VALVE, GATE, WEDGE, OS&Y 4" P150 (4"-P150-CWS-3001/B2) on E-101-N3 | E 113'-6" N -25'-0" EL 109'-3-13/16" | | |
| 6 | blind | E-101-N3 | 4"-P150-CWS-3001 | Fit a blind between 4"-P150-CWS-3001/B2 and the nozzle flange E-101-N3 | E 113'-6" N -25'-0" EL 108'-8-1/16" | | |
| 7 | valve | 4"-P150-CWR-3002/A2 | 4"-P150-CWR-3002 | Close and lock VALVE, GATE, WEDGE, OS&Y 4" P150 (4"-P150-CWR-3002/A2) on E-101-N4 | E 125'-6" N -25'-0" EL 103'-8-3/16" | | |
| 8 | blind | E-101-N4 | 4"-P150-CWR-3002 | Fit a blind between 4"-P150-CWR-3002/A2 and the nozzle flange E-101-N4 | E 125'-6" N -25'-0" EL 104'-3-15/16" | | |
| 9 | valve | 1"-P150-V-6021/2 | 1"-P150-V-6021 | Close and lock VALVE, BALL, FLOATING 1" P150 (1"-P150-V-6021/2) on E-101-N5 | E 124'-6" N -25'-0" EL 109'-1/2" | | |
| 10 | blind | E-101-N5 | 1"-P150-V-6021 | Fit a blind between 1"-P150-V-6021/2 and the nozzle flange E-101-N5 | E 124'-6" N -25'-0" EL 108'-5-11/16" | | |
| 11 | valve | 1"-P150-D-6022/2 | 1"-P150-D-6022 | Close and lock VALVE, BALL, FLOATING 1" P150 (1"-P150-D-6022/2) on E-101-N6 | E 113'-6" N -25'-0" EL 103'-11-1/2" | | |
| 12 | blind | E-101-N6 | 1"-P150-D-6022 | Fit a blind between 1"-P150-D-6022/2 and the nozzle flange E-101-N6 | E 113'-6" N -25'-0" EL 104'-6-3/8" | | |
| 13 | valve | 3/4"-P300-I-6023/2 | 3/4"-P300-I-6023 | Close and lock VALVE, BALL, FLOATING 3/4" P300 (3/4"-P300-I-6023/2) on E-101-N7 | E 119'-6" N -27'-6-1/4" EL 106'-6" | | |
| 14 | blind | E-101-N7 | 3/4"-P300-I-6023 | Fit a blind between 3/4"-P300-I-6023/2 and the nozzle flange E-101-N7 | E 119'-6" N -26'-11-1/2" EL 106'-6" | | |

## Inspections

| Id | Inspection | Criteria | Basis |
|---|---|---|---|
| WO-300-01-I1 | Internal visual examination of the shell, channel and tube bundle | Findings recorded against the owner's integrity criteria | SW-NDE-4 |
| WO-300-01-I2 | Ultrasonic thickness at the 12 TMLs listed | Each reading above the owner's minimum wall | SW-WO-6 |

## Thickness monitoring locations (SW-WO-6)

| TML | Item | Location | Position | Nominal wall | R1 | R2 | R3 | R4 |
|---|---|---|---|---|---|---|---|---|
| WO-300-01-TML01 | E-101 | Nozzle neck E-101-N1 | E 110'-6" N -25'-0" EL 108'-8-1/16" | | top: ____ | side: ____ | bottom: ____ | side: ____ |

| TML | Item | Location | Position | Nominal wall | R1 | R2 | R3 | R4 |
|---|---|---|---|---|---|---|---|---|
| WO-300-01-TML02 | E-101 | Nozzle neck E-101-N2 | E 110'-6" N -25'-0" EL 104'-3-15/16" | | top: ____ | side: ____ | bottom: ____ | side: ____ |
| WO-300-01-TML03 | E-101 | Nozzle neck E-101-N3 | E 113'-6" N -25'-0" EL 108'-8-1/16" | | top: ____ | side: ____ | bottom: ____ | side: ____ |
| WO-300-01-TML04 | E-101 | Nozzle neck E-101-N4 | E 125'-6" N -25'-0" EL 104'-3-15/16" | | top: ____ | side: ____ | bottom: ____ | side: ____ |
| WO-300-01-TML05 | E-101 | Nozzle neck E-101-N5 | E 124'-6" N -25'-0" EL 108'-5-11/16" | | top: ____ | side: ____ | bottom: ____ | side: ____ |
| WO-300-01-TML06 | E-101 | Nozzle neck E-101-N6 | E 113'-6" N -25'-0" EL 104'-6-3/8" | | top: ____ | side: ____ | bottom: ____ | side: ____ |
| WO-300-01-TML07 | E-101 | Nozzle neck E-101-N7 | E 119'-6" N -26'-11-1/2" EL 106'-6" | | top: ____ | side: ____ | bottom: ____ | side: ____ |
| WO-300-01-TML08 | E-101 | Left head centre | E 110'-8-3/4" N -25'-0" EL 106'-6" | | centre: ____ | | | |
| WO-300-01-TML09 | E-101 | Right head centre | E 128'-3-1/4" N -25'-0" EL 106'-6" | | centre: ____ | | | |
| WO-300-01-TML10 | E-101 | Shell at 0.000" from the left tangent | E 111'-6" N -25'-0" EL 106'-6" | | 0°: ____ | 90°: ____ | 180°: ____ | 270°: ____ |
| WO-300-01-TML11 | E-101 | Shell at 6'-0" from the left tangent | E 117'-6" N -25'-0" EL 106'-6" | | 0°: ____ | 90°: ____ | 180°: ____ | 270°: ____ |
| WO-300-01-TML12 | E-101 | Shell at 12'-0" from the left tangent | E 123'-6" N -25'-0" EL 106'-6" | | 0°: ____ | 90°: ____ | 180°: ____ | 270°: ____ |

## Test — Hydrostatic test (tube side)

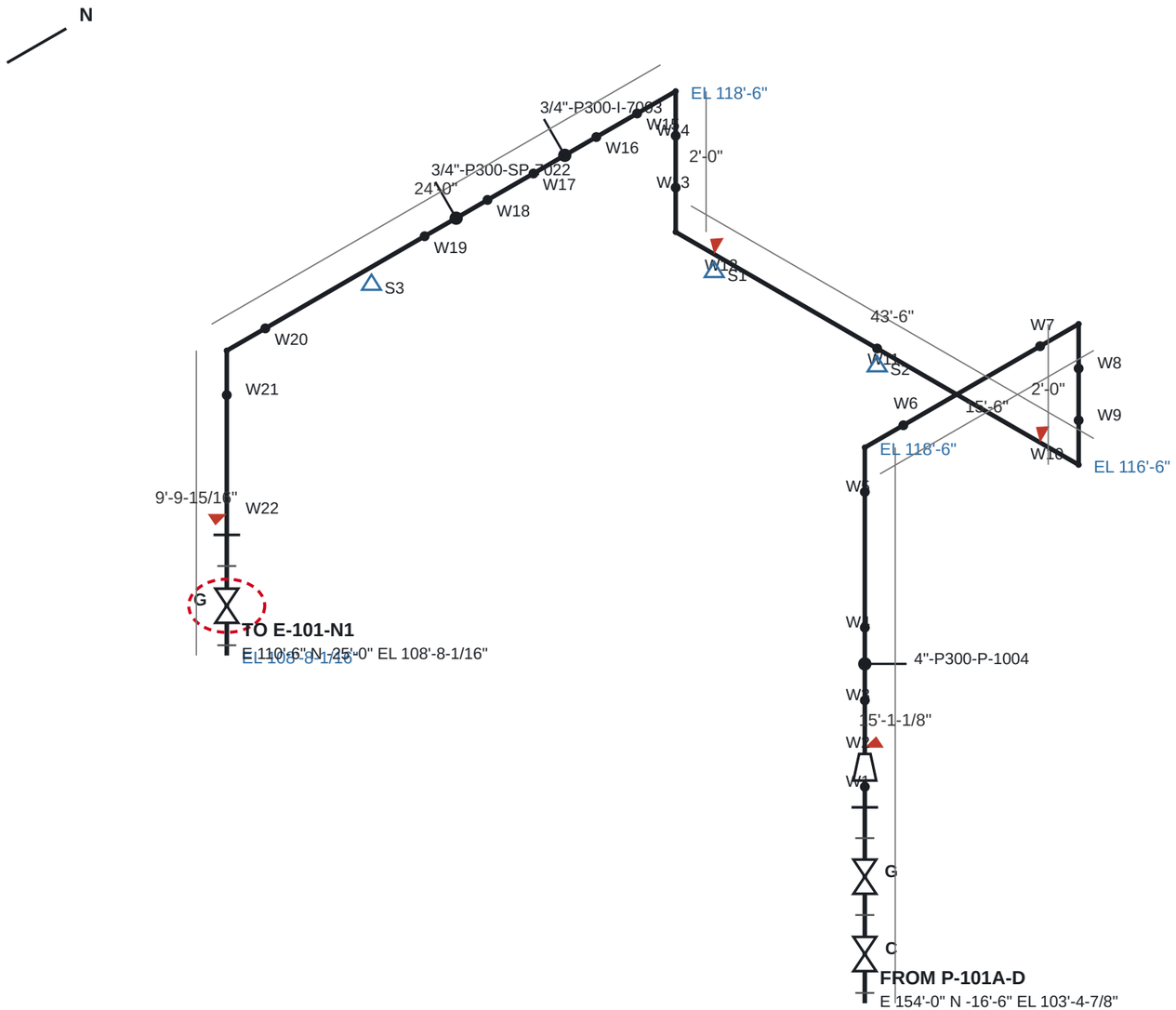
| | |
|---|---|
| **Description** | Tube side of the reinstalled bundle |
| **Pressure** | 420 psi |
| **Hold** | 30 minutes, no leak, no pressure drop |
| **Basis** | SW-TEST-1 |

## Inspection and test plan (SW-ITP-1..3)

H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

| # | Activity | Reference | Acceptance | Record | Contr. | Owner | 3rd party | Status | Signed (contr. / owner / 3rd) | Date | Remark |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained | SW-WO-3, isolation list | Every isolation in place and tagged; zero pressure at the vent | Signed isolation certificate | H | H | — | | | | |
| 2 | Equipment vented, purged and gas-tested for entry; confined-space entry permit issued | Owner entry procedure | Atmosphere safe for entry per owner limits | Gas test record, entry permit | H | H | — | | | | |
| 3 | Channel cover removed; bundle pulled and cleaned | Owner procedure | Surfaces clean enough to inspect | Daily report | S | S | — | | | | |
| 4 | Internal visual examination | SW-NDE-4 | Findings recorded | Inspection report with photographs | W | H | W | | | | |
| 5 | Ultrasonic thickness at 12 TMLs | SW-WO-6 | Above the owner's minimum wall | Thickness record per TML | S | R | R | | | | |
| 6 | Owner's decision on findings (repair, monitor or accept) | Owner integrity procedure | Decision recorded | Signed disposition | R | H | R | | | | |
| 7 | Closed with new gaskets on every opened joint; studs to preload | SW-WO-5 | Preload as stated | Bolt-up record | W | W | — | | | | |
| 8 | Tube-side hydrostatic test at 420 psi, 30 minutes | SW-TEST-1 | No leak, no pressure drop | Test record | H | H | H | | | | |
| 9 | Blinds removed, isolations returned to normal line-up per the isolation list | SW-WO-3 | Every blind removed and accounted for | Isolation certificate closed | H | H | — | | | | |
| 10 | Records complete; work order handed back to operations | SW-ITP-3 | All ITP activities accepted or not applicable | Signed ITP and completion record | H | H | R | | | | |

**ISOMETRIC — 4"-P300-P-1003 — Pull, clean and test the tube bundle of E-101**




## BILL OF MATERIALS

| | | |
|---|---|---|
| 1 | GASKET 3" P300 COMPRESSED SHEET | 3 |
| 2 | VALVE, SWING CHECK 3" P300 | 1 |
| 3 | STUD SET 4 × 0.875" × 5-3/4" WITH 2 NUTS, FOR 3" P300 | 3 |
| 4 | VALVE, GATE, WEDGE, OS&Y 3" P300 | 1 |
| 5 | FLANGE WELD-NECK 3" P300 RAISED FACE | 1 |
| 6 | REDUCER CONCENTRIC 4" × 3" P300 | 1 |
| 7 | PIPE 4" P300 BEVELLED ENDS | 98'-8-7/16" |
| 8 | TEE EQUAL 4" P300 | 1 |
| 9 | ELBOW 90° R = 5/3 OD 4" P300 | 6 |
| 10 | TEE REDUCING 4" × 3/4" P300 | 2 |
| 11 | STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P300 | 2 |
| 12 | FLANGE WELD-NECK 4" P300 RAISED FACE | 1 |
| 13 | GASKET 4" P300 COMPRESSED SHEET | 2 |
| 14 | VALVE, GATE, WEDGE, OS&Y 4" P300 | 1 |

## WELDS 22 (4 FIELD) · SUPPORTS 3 · SPOOLS 5

W1 W2F W3 W4 W5 W6 W7 W8 W9 W10F
W11 W12F W13 W14 W15 W16 W17 W18 W19 W20
W21 W22F

## 4"-P300-P-1003

CLASS H300 — Hydrocarbon process
SERVICE P · 4" · P300 · DESIGN 300 psi
P-101A-D → E-101-N1
PIPE 98'-8-7/16" · WEIGHT 1276 lb

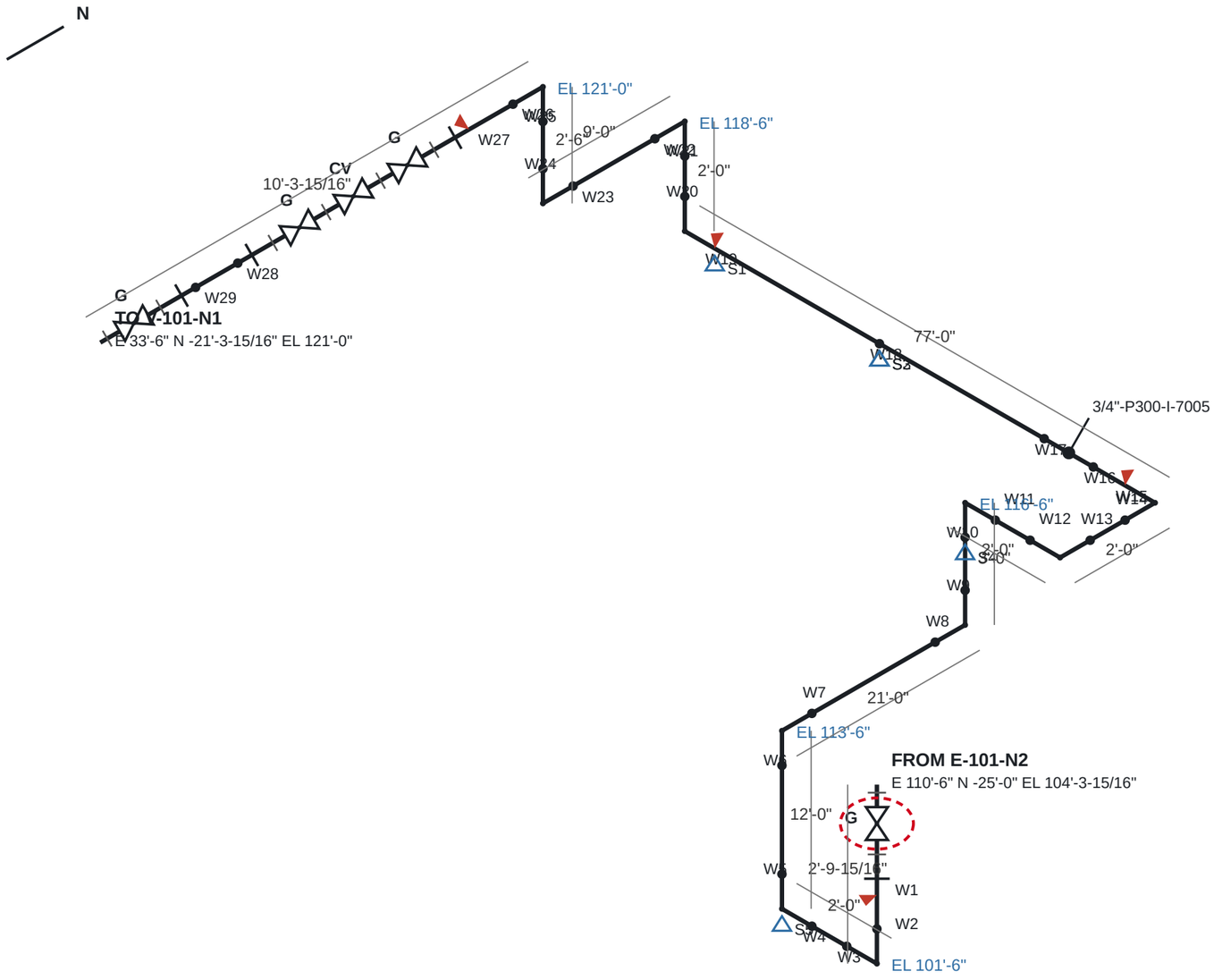
**DWG ISO-4"-P300-P-1003 REV 0 WO-300-01**

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## Isometric ISO-4"-P300-P-1005

ISOMETRIC — 4"-P300-P-1005 — Pull, clean and test the tube bundle of E-101




## BILL OF MATERIALS

| | | |
|---|---|---|
| 1 | GASKET 4" P300 COMPRESSED SHEET | 8 |
| 2 | VALVE, GATE, WEDGE, OS&Y 4" P300 | 4 |
| 3 | STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P300 | 8 |
| 4 | FLANGE WELD-NECK 4" P300 RAISED FACE | 4 |
| 5 | PIPE 4" P300 BEVELLED ENDS | 124'-6-1/2" |
| 6 | ELBOW 90° R = 5/3 OD 4" P300 | 11 |
| 7 | TEE REDUCING 4" × 3/4" P300 | 1 |
| 8 | VALVE, CONTROL, GLOBE WITH DIAPHRAGM ACTUATOR 4" P300 | 1 |

## WELDS 29 (4 FIELD) · SUPPORTS 5 · SPOOLS 6

W1F W2 W3 W4 W5 W6 W7 W8 W9 W10
W11 W12 W13 W14 W15F W16 W17 W18 W19F W20
W21 W22 W23 W24 W25 W26 W27F W28 W29

## 4"-P300-P-1005

CLASS H300 — Hydrocarbon process
SERVICE P · 4" · P300 · DESIGN 300 psi
E-101-N2 → V-101-N1
PIPE 124'-6-1/2" · WEIGHT 1974 lb

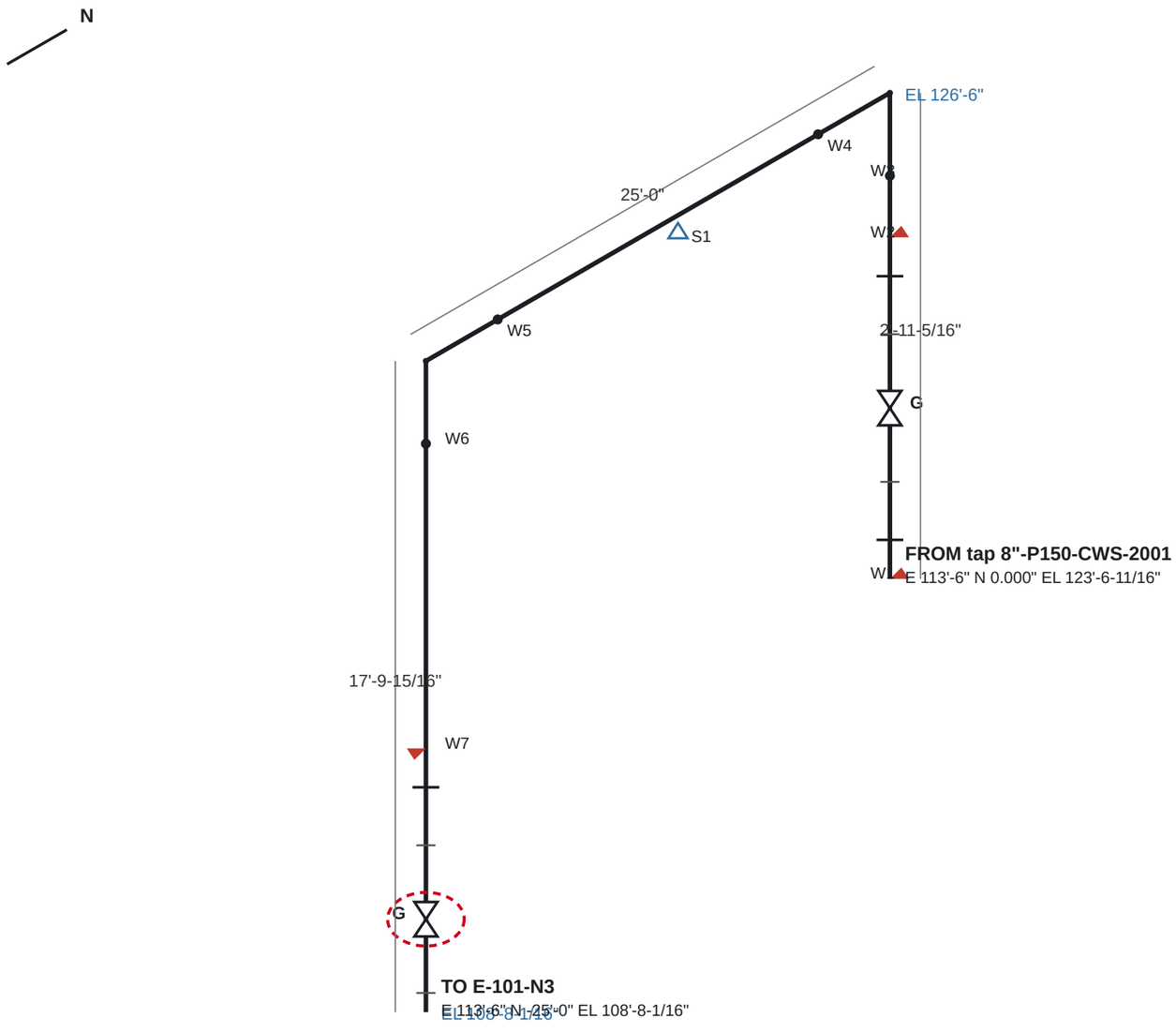
**DWG ISO-4"-P300-P-1005 REV 0 WO-300-01**

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## Isometric ISO-4"-P150-CWS-3001

**ISOMETRIC — 4"-P150-CWS-3001 — Pull, clean and test the tube bundle of E-101**




## BILL OF MATERIALS

| | | |
|---|---|---|
| 1 | FLANGE WELD-NECK 4" P150 RAISED FACE | 3 |
| 2 | GASKET 4" P150 COMPRESSED SHEET | 4 |
| 3 | VALVE, GATE, WEDGE, OS&Y 4" P150 | 2 |
| 4 | STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P150 | 4 |
| 5 | PIPE 4" P150 BEVELLED ENDS | 39'-7-5/16" |
| 6 | ELBOW 90° R = 5/3 OD 4" P150 | 2 |

**WELDS 7 (3 FIELD) · SUPPORTS 1 · SPOOLS 4**

W1F W2F W3 W4 W5 W6 W7F

## 4"-P150-CWS-3001

CLASS W150 — Water (cooling and process)
SERVICE CWS · 4" · P150 · DESIGN 150 psi
tap 8"-P150-CWS-2001 → E-101-N3
PIPE 39'-7-5/16" · WEIGHT 693 lb

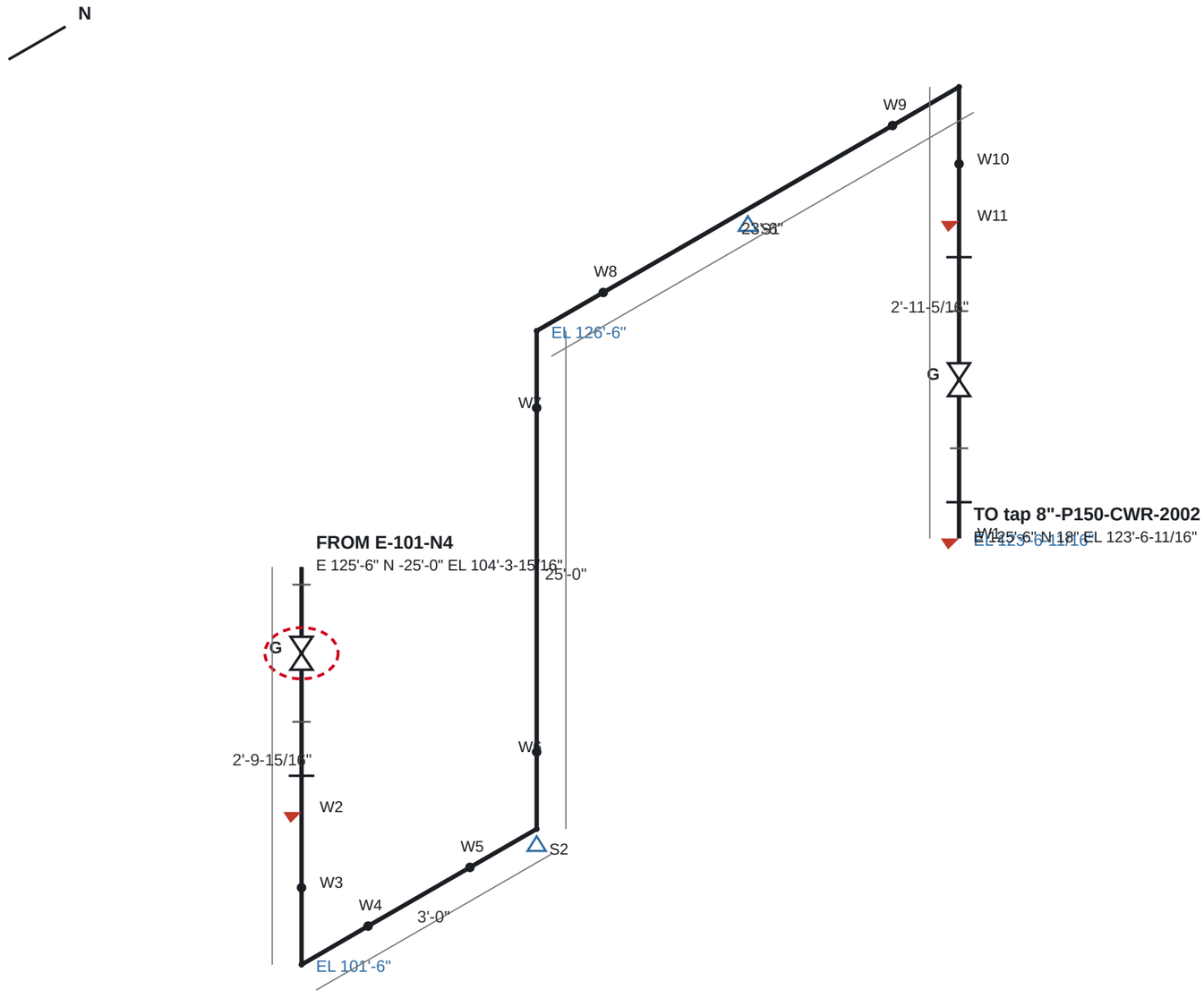
**DWG ISO-4"-P150-CWS-3001 REV 0 WO-300-01**

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## Isometric ISO-4"-P150-CWR-3002

**ISOMETRIC — 4"-P150-CWR-3002 — Pull, clean and test the tube bundle of E-101**




## BILL OF MATERIALS

| | | |
|---|---|---|
| 1 | GASKET 4" P150 COMPRESSED SHEET | 4 |
| 2 | VALVE, GATE, WEDGE, OS&Y 4" P150 | 2 |
| 3 | STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P150 | 4 |
| 4 | FLANGE WELD-NECK 4" P150 RAISED FACE | 3 |
| 5 | PIPE 4" P150 BEVELLED ENDS | 48'-10-5/8" |
| 6 | ELBOW 90° R = 5/3 OD 4" P150 | 4 |

**WELDS 11 (3 FIELD) · SUPPORTS 2 · SPOOLS 4**

W1F W2F W3 W4 W5 W6 W7 W8 W9 W10
W11F

## 4"-P150-CWR-3002

CLASS W150 — Water (cooling and process)
SERVICE CWR · 4" · P150 · DESIGN 150 psi
E-101-N4 → tap 8"-P150-CWR-2002
PIPE 48'-10-5/8" · WEIGHT 777 lb

**DWG ISO-4"-P150-CWR-3002 REV 0 WO-300-01**

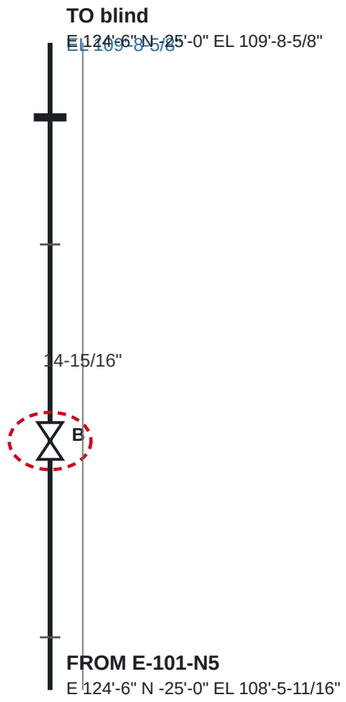
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## Isometric ISO-1"-P150-V-6021

ISOMETRIC — 1"-P150-V-6021 — Pull, clean and test the tube bundle of E-101

N




## BILL OF MATERIALS

| | | |
|---|---|---|
| 1 | GASKET 1" P150 COMPRESSED SHEET | 2 |
| 2 | VALVE, BALL, FLOATING 1" P150 | 1 |
| 3 | STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150 | 2 |
| 4 | FLANGE BLIND 1" P150 RAISED FACE | 1 |

**WELDS 0 (0 FIELD) · SUPPORTS 0 · SPOOLS 0**

## 1"-P150-V-6021

CLASS H150 — Hydrocarbon process
SERVICE V · 1" · P150 · DESIGN 150 psi
E-101-N5 → blind
PIPE 0.000" · WEIGHT 39 lb

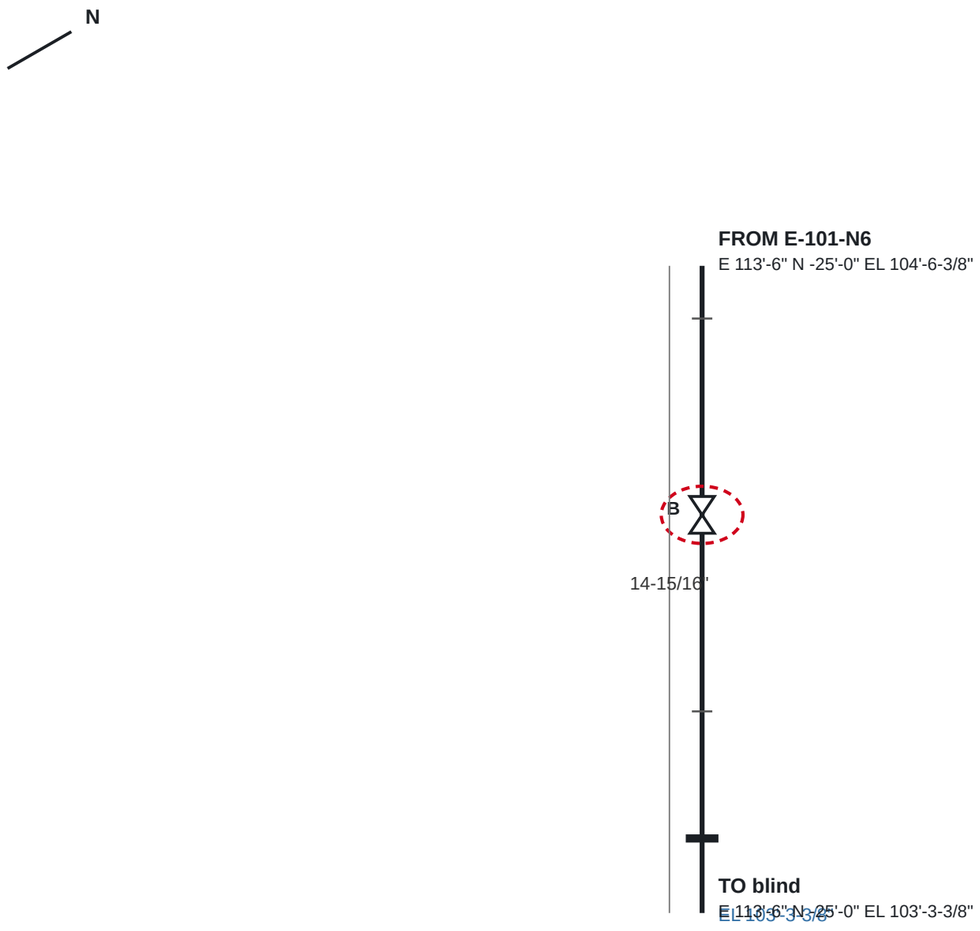
**DWG ISO-1"-P150-V-6021 REV 0 WO-300-01**

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## Isometric ISO-1"-P150-D-6022

**ISOMETRIC — 1"-P150-D-6022 — Pull, clean and test the tube bundle of E-101**




**BILL OF MATERIALS**

| | | |
|---|---|---|
| 1 | GASKET 1" P150 COMPRESSED SHEET | 2 |
| 2 | VALVE, BALL, FLOATING 1" P150 | 1 |
| 3 | STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150 | 2 |
| 4 | FLANGE BLIND 1" P150 RAISED FACE | 1 |

**WELDS 0 (0 FIELD) · SUPPORTS 0 · SPOOLS 0**

## 1"-P150-D-6022

CLASS H150 — Hydrocarbon process
SERVICE D · 1" · P150 · DESIGN 150 psi
E-101-N6 → blind
PIPE 0.000" · WEIGHT 39 lb

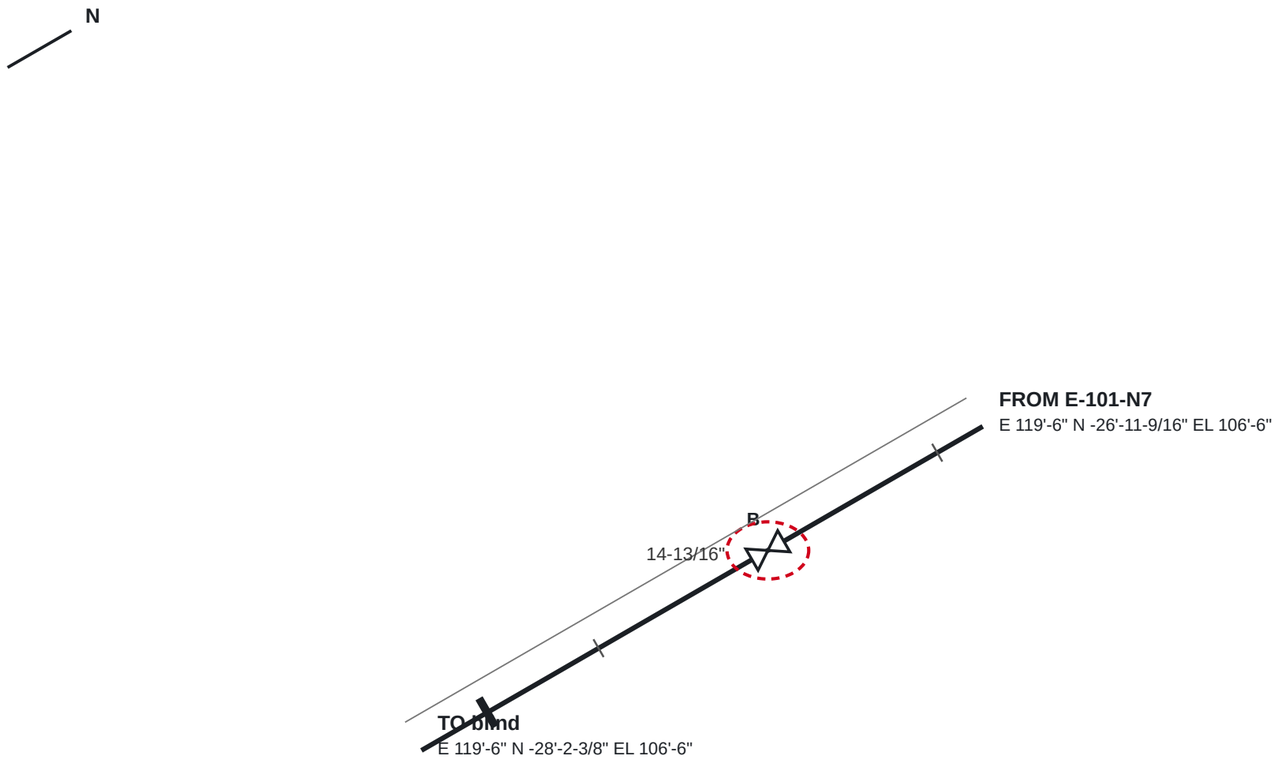
**DWG ISO-1"-P150-D-6022 REV 0 WO-300-01**

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## Isometric ISO-3/4"-P300-I-6023

**ISOMETRIC — 3/4"-P300-I-6023 — Pull, clean and test the tube bundle of E-101**




## BILL OF MATERIALS

| | | |
|---|---|---|
| 1 | GASKET 3/4" P300 COMPRESSED SHEET | 2 |
| 2 | VALVE, BALL, FLOATING 3/4" P300 | 1 |
| 3 | STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P300 | 2 |
| 4 | FLANGE BLIND 3/4" P300 RAISED FACE | 1 |

**WELDS 0 (0 FIELD) · SUPPORTS 0 · SPOOLS 0**

## 3/4"-P300-I-6023

CLASS H300 — Hydrocarbon process
SERVICE I · 0.75" · P300 · DESIGN 300 psi
E-101-N7 → blind
PIPE 0.000" · WEIGHT 32 lb

**DWG ISO-3/4"-P300-I-6023 REV 0 WO-300-01**

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

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